

# Ghali Omar Boutaib

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## Education

**University of South Florida**, BS in Computer Engineering, Tampa FL Aug 2022 – May 2026

- **GPA:** Cumulative: 3.91/4.0, Major: 3.92/4.0
- **Advanced Coursework:** Computer Architecture, CMOS/VLSI Design, Electrical Systems, Electronic Materials, Operating Systems, Hardware Accelerators for ML, Cryptographic Hardware and Embedded Systems, etc.
- **Research:** SEES Lab, Bellini College of Artificial Intelligence, Cybersecurity and Computing

## Research experience

### Past: Veribest Dataset and Pyranet Paper

Gathered and organized millions of samples of Verilog code for the development of an award winning dataset for LLM fine-tuning. Presented findings at ICCAD conference and earned awards from NVIDIA and the NSF. Contributed to writing the paper presenting our findings, showing considerable fine-tuning value. Paper presented at DAC 2025.

### Verimind Paper

Created a self checking LLM framework for Verilog RTL code writing, showing considerable success. Developed a novel evaluation metric building on pass@k to fairly and accurately evaluate frameworks that implement a feedback loop.

### Current: Novel Verilog code generation benchmark

Developing a novel benchmark dataset and framework for the evaluation of large language models on realistic Verilog RTL design tasks.

## Publications and Presentations

**PyraNet: A Large Scale Hierarchical Verilog Dataset** Jun 2025

Bardia Nadimi, **Ghali Omar Boutaib**, Hao Zheng

Proceedings of the 2025 62nd ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, USA, 2025

DOI: 10.1109/DAC63849.2025.11133406 (Earlier version: 10.48550/arXiv.2412.06947, Dec 2024)

**VeriMind: Agentic LLM for Automated Verilog Generation with a Novel Evaluation Metric** Mar 2025

Bardia Nadimi, **Ghali Omar Boutaib**, Hao Zheng

DOI: 10.48550/arXiv.2503.16514

## Awards and Honors

**Veribest Dataset** ICCAD 2024

- Developed the first-place winning dataset for the LLM4HWDesign contest at the 2024 International Conference for Computer-Aided Design.
- Our data set was approximately 50x larger than the second place, scoring the highest on the contest benchmark.
- Presented process and results at the conference.
- Awarded a GPU from NVIDIA corp. and a cash prize from the National Science Foundation.

**Scholarships** Florida Bright Futures, Sanford & Gertrude Hauber Scholarship, Terrell & Neva Sessums Scholarship, Judy Genshaft Honors College Scholarship, CRA UR2PhD FHTC Award

## Professional Experience

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**Research Assistant**, SEES Lab – College of Engineering, USF Tampa May 2024 – Current

- Worked under Dr. Hao Zheng, alongside Ph.D. student Bardia Nadimi in research.
- Won 1st place at the LLM4HWDDesign contest, and presented at ICCAD 2024. Earned prizes from NVIDIA and the NSF. Published a paper presented at DAC 2025.

**Freelance software engineer**, Remote Jan 2023 – Mar 2024

- **AMB Innovations:** Worked with AMB Innovations LLC to develop a Platform as a Service for the food industry. Spearheaded the development of both back and front-end and successfully deployed Beta version using cloud services.
- **USF Biology Department:** Developed an online directory platform for the Phages research program. Allowed students, faculty, guest researches, and prospective students to share contact information and crucial information and news.

**Software Engineering Intern**, Spacecrate.io – Remote Summer 2020

- **Work under senior engineers:** Was supervised by Senior Engineers at the company and worked on the software behind website builder platform. Reinforced soft skills in corporate communication and technical skills in code collaboration.
- **Assigned solo projects:** Developed a media converter tool for customers under the guidance of senior engineers and in collaboration with other interns.

## Humanitarian Work

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Contributed extensive humanitarian and social work in Morocco. All done with LCC7 NGO in Casablanca, Morocco.

- **Ffour LCC7 (5 years):** Served meals for around a thousand people every night during the month of Ramadan.
- **Warmth for all (Dec 2019):** Traveled to the remote region of Imindounit with volunteers and NGO leadership to deliver two trucks of warm clothing and school supplies.
- **Operation cataract (Feb 2020):** Assisted in the registration and treatment of dozens of cataract patients, recruited volunteer surgeons and nurses, and solicited donations of medical equipment.

## Leadership and personal development

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**CRA UR2PhD:** Participating in the CRA UR2PhD research course. Recipient of FHTC award through USF.

**US Universities Arabic Debate Championship:** Represented USF at the 5th edition of the competition against 39 national teams. Secured 12th place overall, won two "best speaker" awards for excellence in argumentation. [[usf.edu/honors/news/2024/event-recap-arabic-debate-competition.aspx](https://usf.edu/honors/news/2024/event-recap-arabic-debate-competition.aspx)]

**Study Abroad: USF Honors Germany Innovation in Engineering:** Engaged in a six-week immersive program focused on innovation, sustainability, and engineering advancements in Osnabrueck, Germany. Participated in hands-on lab practicals, site visits to leading engineering facilities, and collaborative international projects.

**Director of the Raven:** Serve as the Director for the Raven. The USF Honors College's Raven's Guild undergraduate journal. Manages the editorial process. First edition set to publish during Spring 2026.

**Professional photographer:** Portfolio available online [[portfolio.ghaliomar.com](https://portfolio.ghaliomar.com)]

## Other relevant skills

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**Programming languages:** C++, C, Javascript, Python (Earned Certificate), Rust

**Relevant Technologies:** Vivado (Verilog and other HDL), Cloud Infrastructures (Earned Certificate), Linux, Systems Administration, Full-stack platform development,  $\LaTeX$

**Fluently spoken languages:** English, French, Spanish, M.S. Arabic, Moroccan Darija